



**Empowering Future
Tech Professionals**

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TASK BOOKLET – DATA SCIENCE

Program: EncoderX Remote Internship Batch 02

Track: Data Science

Week: 04

★
LEARN
DESIGN
CREATE
INSPIRE

★ ★ ★

ABOUT ENCODERX

EncoderX provides industry-oriented remote internship experiences designed to help students build practical skills, develop professional portfolios, and gain real-world project exposure.

DO'S

- ✓ Maintain professional communication.
- ✓ Use GitHub for version control.
- ✓ Document your work clearly.
- ✓ Follow coding standards and best practices.
- ✓ Submit before deadlines.

DON'TS

- ✗ Do not submit copied projects.
- ✗ Do not share confidential project material.
- ✗ Avoid incomplete documentation.
- ✗ Do not falsify results or reports.

TASK 4: INTERACTIVE DASH- BOARD DEVELOPMENT

W

EXPECTED DURATION

3–5 Days

D

DIFFICULTY LEVEL

Beginner to Intermediate

R

PRIMARY AREA

Data Visualization

F

DOMAIN

Data Science

OBJECTIVE

The objective of this task is to design and develop an interactive business dashboard that enables stakeholders to monitor key performance indicators (KPIs), track trends, and make informed decisions through data visualization.

TASK OVERVIEW

Business dashboards play a critical role in transforming raw data into actionable insights.

In this task, interns will create a professional dashboard using Power BI, Tableau, or Streamlit that presents important metrics through interactive charts, filters, and reports.

The final dashboard should provide a user-friendly way to monitor business performance, explore trends, and understand important insights from the selected dataset.

LEARNING OUTCOMES

Upon successful completion of this task, interns will be able to:

- ✓ Design professional business dashboards.
- ✓ Create interactive visualizations.
- ✓ Implement KPI monitoring systems.
- ✓ Improve data storytelling and reporting skills.
- ✓ Present insights in a user-friendly format.

INSTRUCTIONS

1

Step 1: Dataset Selection

Select a business-related dataset for your dashboard.
You may choose a dataset such as:

- Sales Data
- Marketing Analytics
- Financial Records
- Customer Data
- HR Analytics

Choose a dataset that contains enough meaningful information to support KPI analysis, trends, comparisons, and interactive visualizations.

2

Step 2: KPI Identification

Identify at least five relevant Key Performance Indicators (KPIs) based on your selected dataset.
Examples include:

- Total Revenue
- Profit Margin
- Customer Growth
- Conversion Rate
- Monthly Sales Trends

The selected KPIs should help stakeholders understand important aspects of business performance.

3

Step 3: Dashboard Design

Design a professional and user-friendly dashboard.
Your dashboard should include:

- KPI Summary Cards
- Interactive Charts
- Trend Analysis
- Filters and Slicers
- Comparative Analysis

Organize the dashboard clearly so users can quickly understand the most important metrics and insights.

4**Step 4: Development**

Build the interactive dashboard using one of the following tools:

- Power BI
- Tableau
- Streamlit

Implement the selected KPIs, visualizations, filters, trend analysis, and comparative analysis within the dashboard.

5**Step 5: Deployment**

Make the completed dashboard accessible through a shareable link.

If using Streamlit, deploy the application online.

For Power BI or Tableau, publish the dashboard and generate a shareable link.

Verify that the dashboard can be accessed and viewed successfully.

6**Step 6: Documentation**

Prepare clear project documentation explaining the dashboard and the insights generated.

The documentation should explain:

- Dashboard features
- Selected KPIs
- Important visualizations
- Key insights generated from the data

The documentation should help another person understand the purpose, functionality, and findings of the dashboard.

DELIVERABLES

- ✓ Dashboard File
- ✓ Deployment / Sharing Link
- ✓ Project Documentation

- ✓ Dashboard Screenshots
- ✓ GitHub Repository Link

EVALUATION CRITERIA

Criteria	Weightage
Dashboard Design	25%
KPI Selection	20%
Interactivity & Usability	25%
Insights & Reporting	15%
Documentation	15%

TASK INFORMATION

- ✓ **Task:** Interactive Dashboard Development
- ✓ **Domain:** Data Science
- ✓ **Week:** 04
- ✓ **Expected Duration:** 3–5 Days
- ✓ **Difficulty Level:** Beginner to Intermediate
- ✓ **Primary Area:** Data Visualization
- ✓ **Recommended Technologies:** Power BI, Tableau, Streamlit
- ✓ **Core Skills:** Dashboard Design, KPI Monitoring, Interactive Visualization, Data Storytelling, Trend Analysis, Comparative Analysis, Reporting

SUBMISSION REQUIREMENTS

- ✓ Upload all project files to a public GitHub repository.
- ✓ Maintain a clear and professional README file.
- ✓ Record a 3–5 minute project demonstration video.
- ✓ Publish a LinkedIn post describing your project and key findings.
- ✓ Tag EncoderX in the LinkedIn post.
- ✓ Include the hashtags: [#EncoderX](#) [#DataScience](#) [#Internship](#) [#LearningInPublic](#)
- ✓ Submit the GitHub Repository Link, LinkedIn Post Link, and Video Link in a single PDF through the assigned submission portal.

FINAL SUBMISSION CHECKLIST

- ✓ Business-related dataset selected.
- ✓ At least five relevant KPIs identified.
- ✓ KPI Summary Cards implemented.
- ✓ Interactive charts created.
- ✓ Trend analysis included.
- ✓ Filters and slicers implemented.
- ✓ Comparative analysis included.

- ✓ Dashboard developed using Power BI, Tableau, or Streamlit.
- ✓ Dashboard published or deployed with a shareable link.
- ✓ Dashboard features documented.
- ✓ KPIs and generated insights documented.
- ✓ Dashboard screenshots included.
- ✓ Clean source files uploaded to GitHub.
- ✓ Professional README included.
- ✓ 3–5 minute project demonstration video recorded.
- ✓ LinkedIn post published and EncoderX tagged.
- ✓ Required links submitted in a single PDF.

ALL THE BEST!

Build an interactive business dashboard, transform data into actionable insights, and demonstrate your data storytelling skills.